

Control

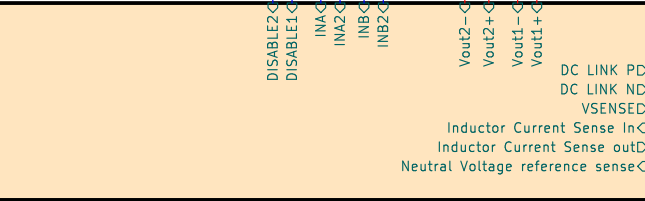


File: St's.controlpipeline.kicad_sch

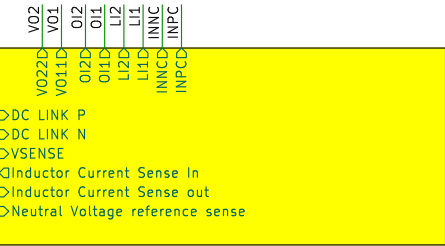
DC Power Converter



File: power.kicad_sch



File: switching.kicad_sch



File: out_sense.kicad_sch

Sensing Elements

Inverter Mosfets

Navrachana University

Sheet: /
File: AntiVirus.kicad_sch

Title:

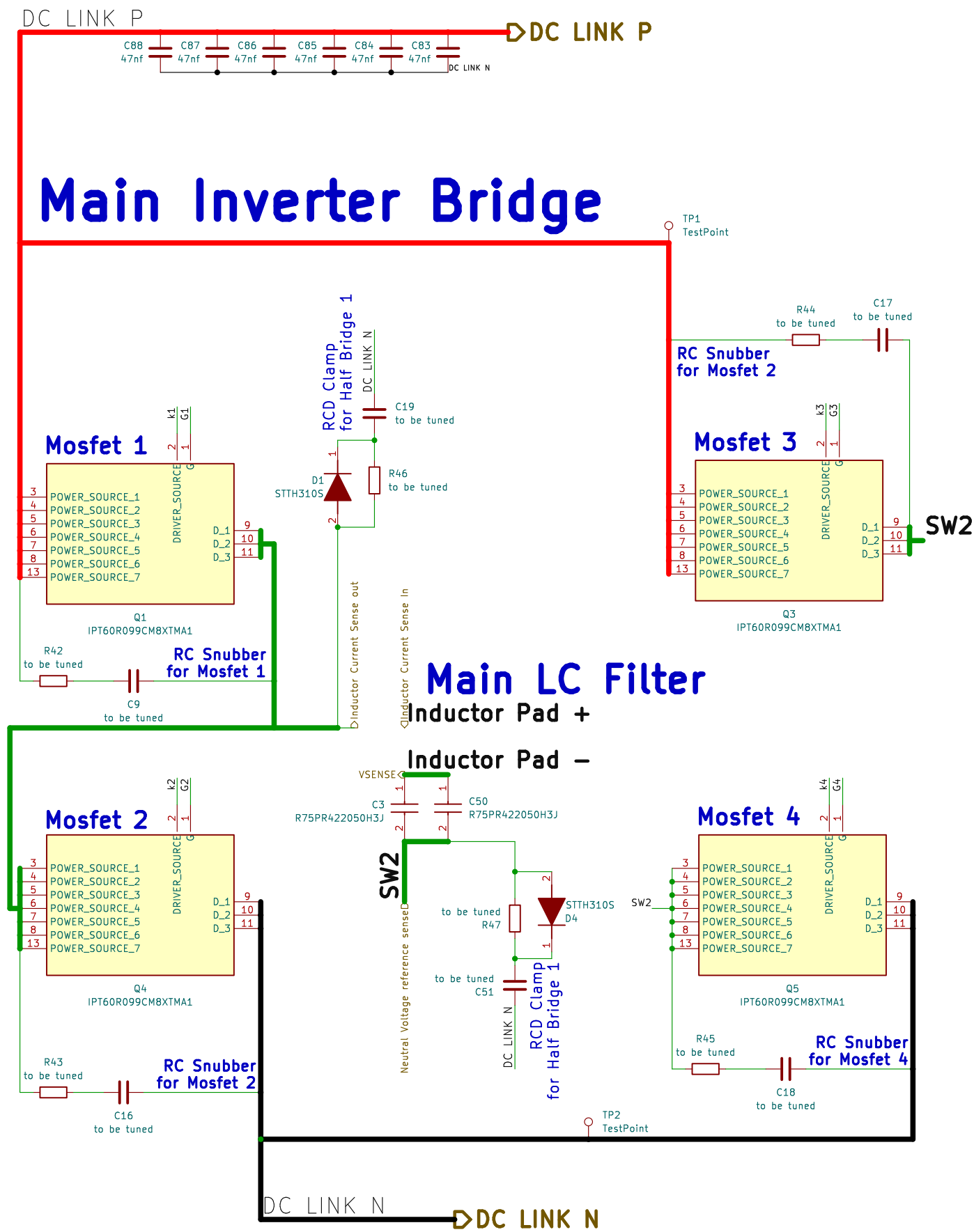
Size: A3 Date: 2026-04-21

Rev: 1

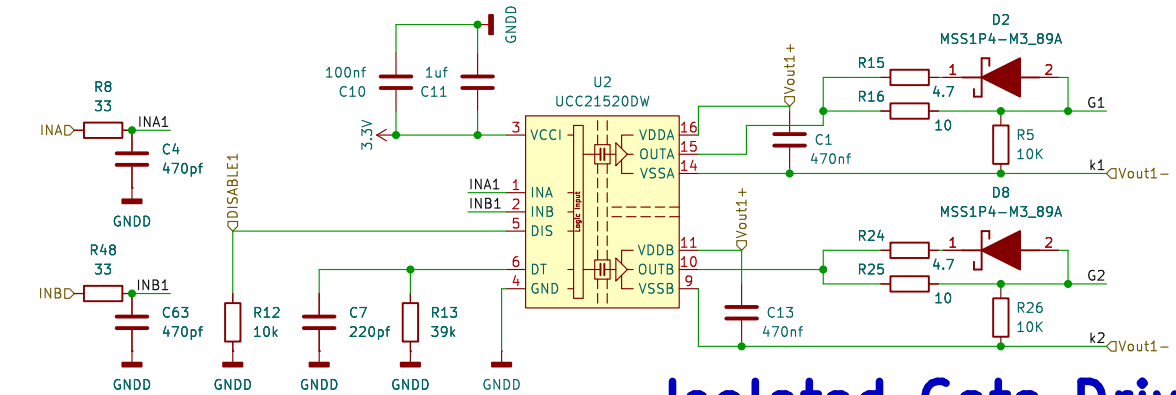
KiCad E.D.A. 9.0.0

Id: 1/5

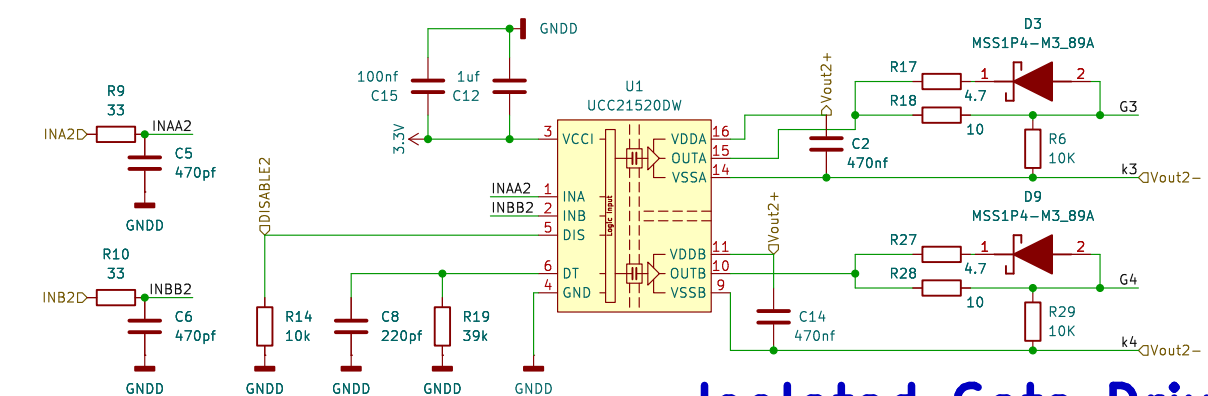
Main Inverter Bridge



Isolated Gate Driver for Mosfet 1 and Mosfet 2

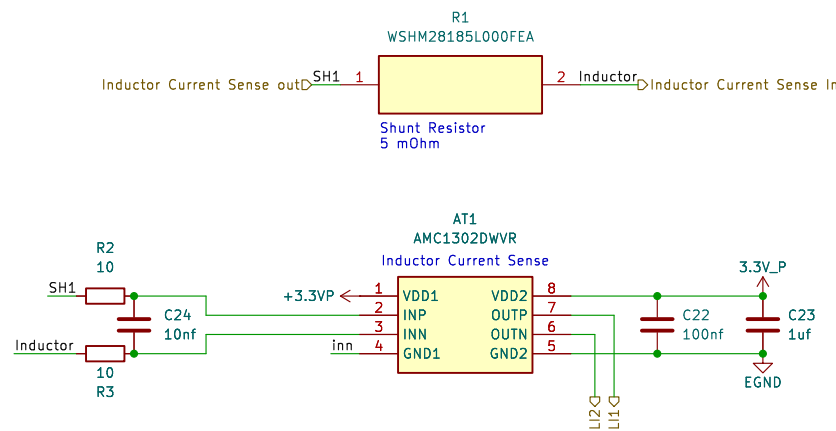


Isolated Gate Driver for Mosfet 3 and mosfet 4



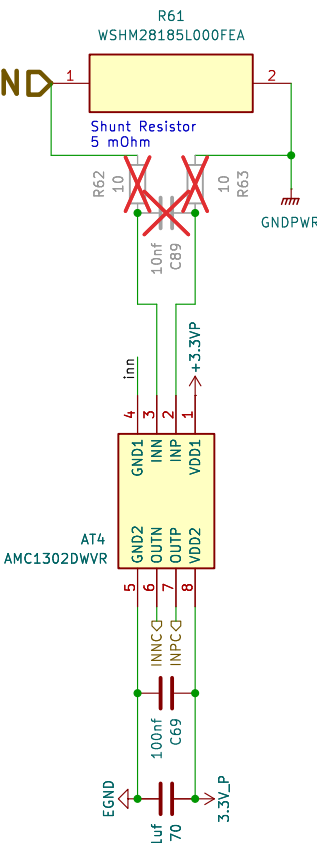
Output LC Filter Inductor Measurement

Using shunt and Isolated current sense voltage amplifier



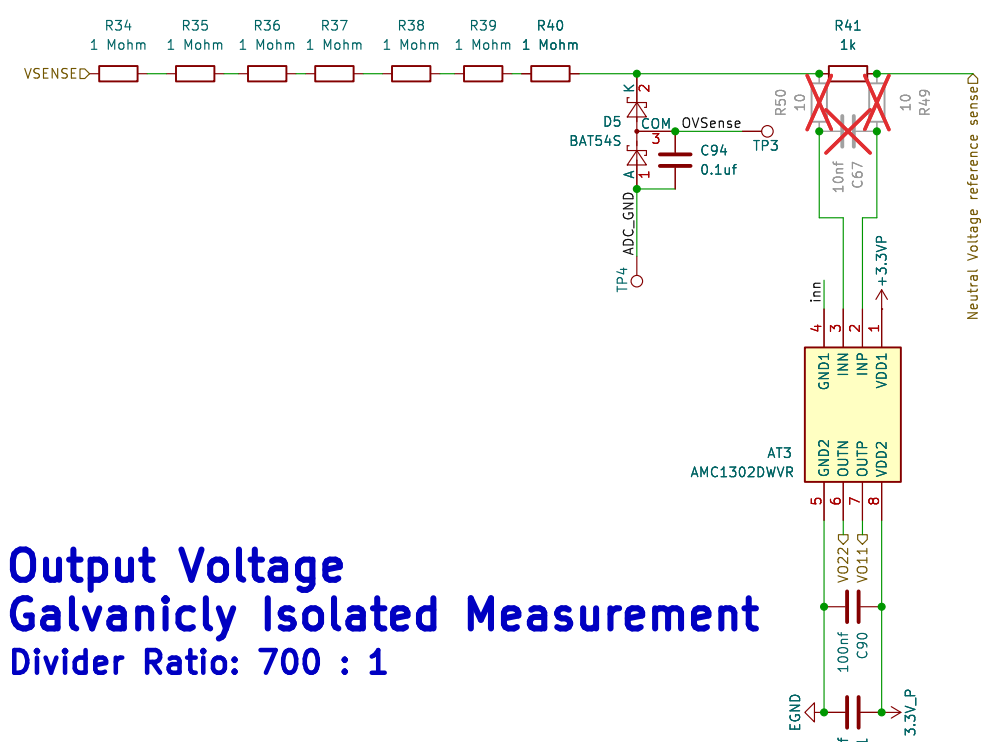
DC LINK ND

Input Current Mesurement
Using shunt and Isolated current sense voltage amplifier

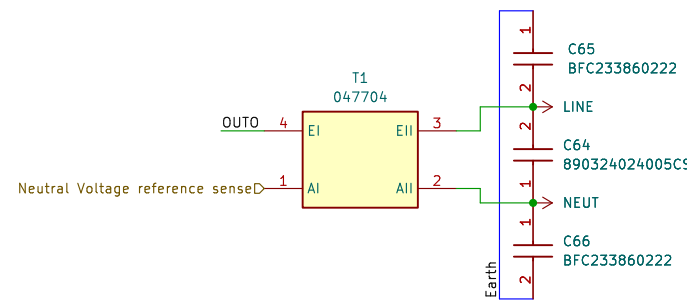


Output Voltage Galvanicly Isolated Measurement

Divider Ratio: 700 : 1

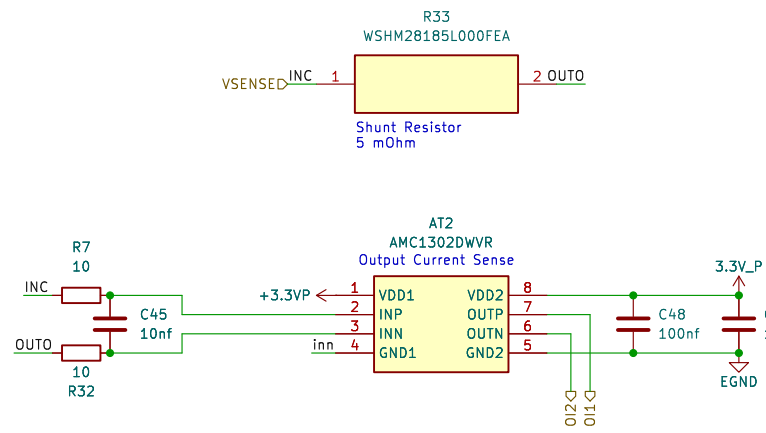


Output EMI Filter



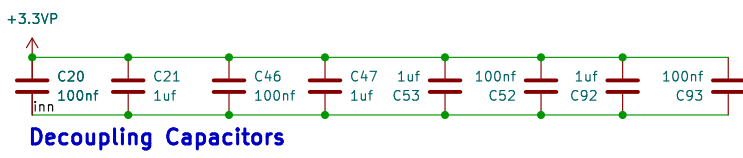
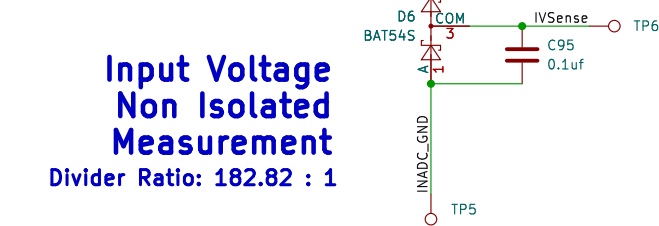
Output Current Measurement

Using shunt and Isolated current sense voltage amplifier

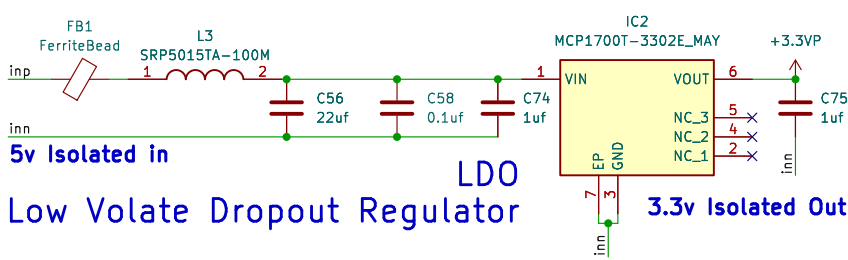


DC LINK PD

Input Voltage Non Isolated Measurement
Divider Ratio: 182.82 : 1

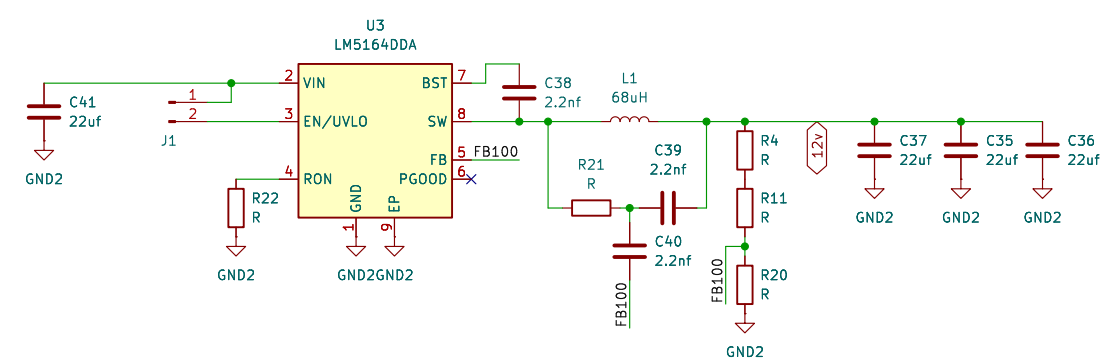


Decoupling Capacitors

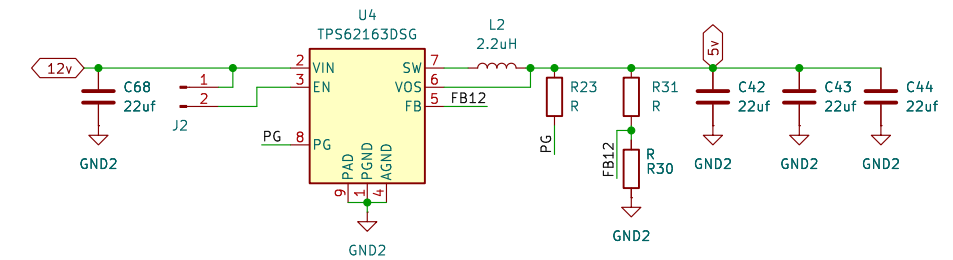


5v Isolated in
LDO
Low Volate Dropout Regulator
3.3v Isolated Out

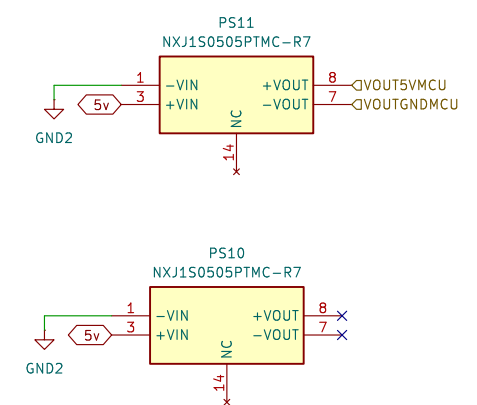
Vin to 12v converter
using LM5164DDA



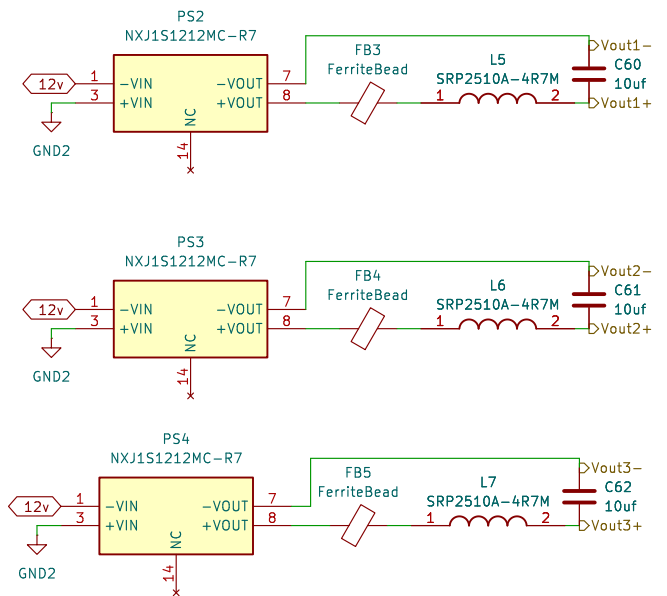
12v to 5v converter
using TPS62163DSG



12v isolaters for Gate power supply
using NXJ1S1212MC-R7



12v isolaters for Gate power supply
using NXJ1S1212MC-R7



TPS62163DSG converts 12v to 5v
LM5164DDA converts Vin to 12V
Vin Nominal input of 48v and max 80v

Navrachana University

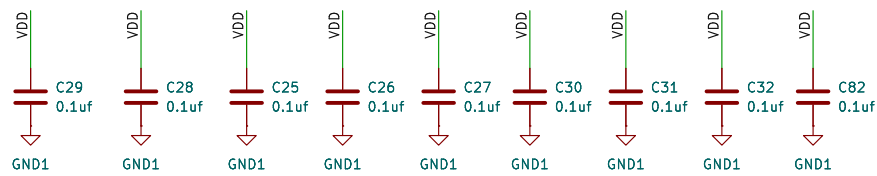
Sheet: /DC Power Converter/
File: power.kicad_sch

Title: Power conversion for MCU and Gate Driver

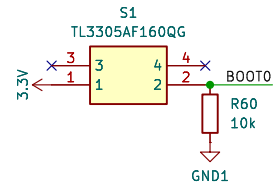
Size: A3 Date: 2026-04-21
KiCad E.D.A. 9.0.0

Rev: 1
Id: 4/5

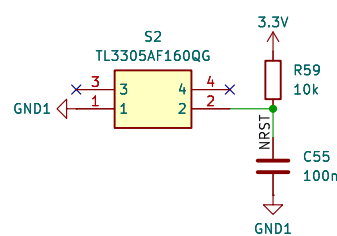
MCU Decoupling Capacitor



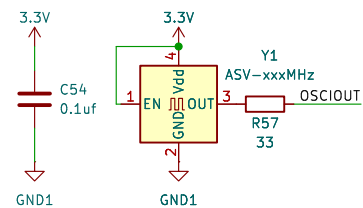
Boot Button



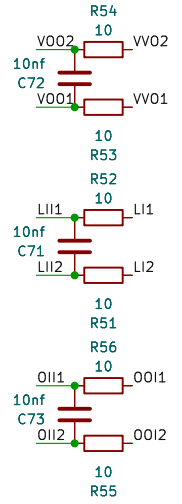
Reset Button



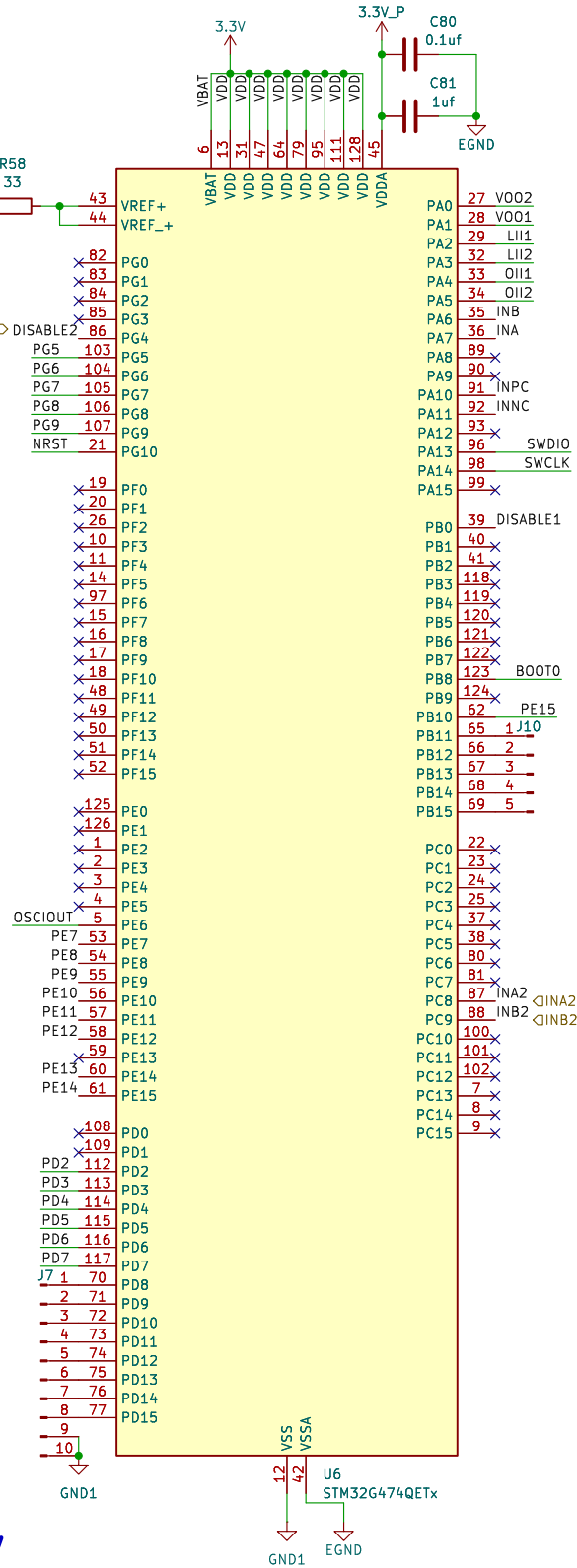
8 MHz Oscillator



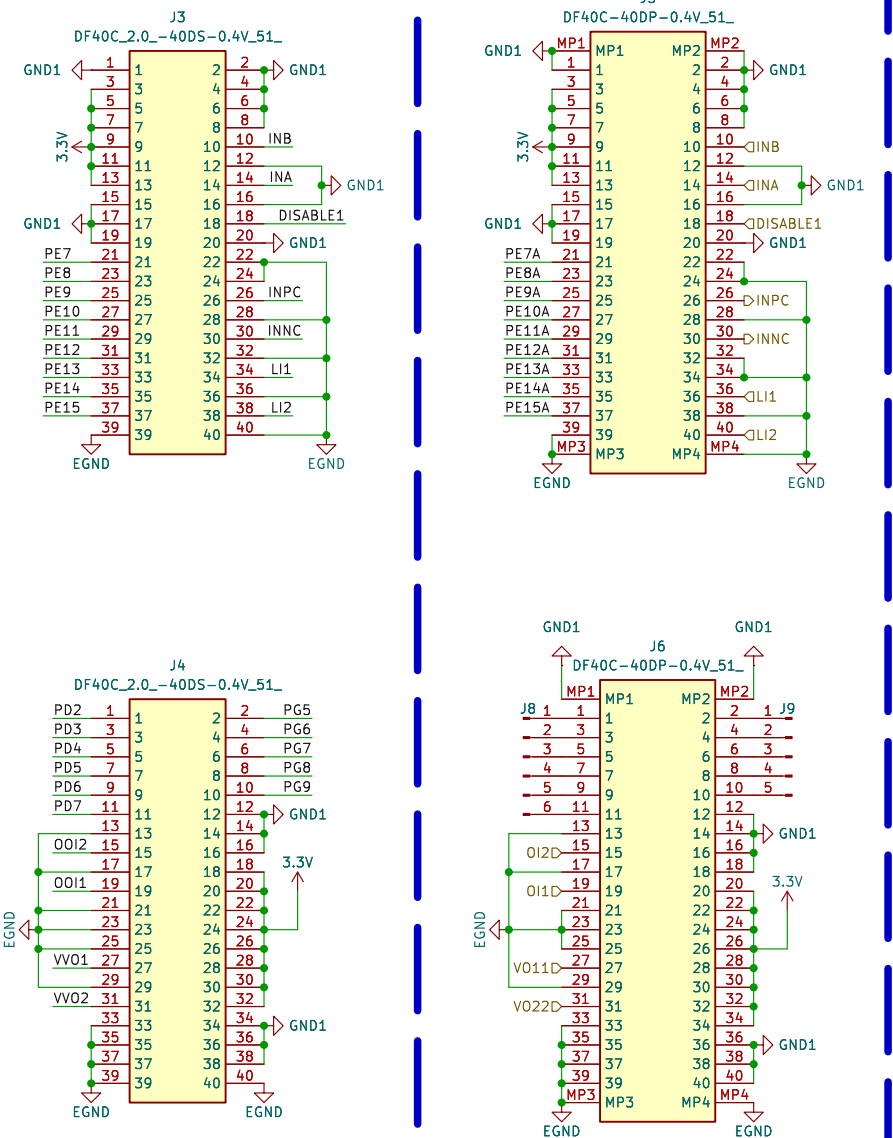
Differential Sensing Input Filter



Main MCU = STM32G474QET3



MCU Board to Main Board Connector



3.3v Out for Digital section only

3.3v Out for Analog section only

STM32G474QET3 is being used as main MCU

Navrachana University

Sheet: /Control/

File: St's.controlpipeline.kicad_sch

Title: Microcontroller

Size: A3

Date: 2026-04-21

Rev: 1

KiCad E.D.A. 9.0.0

Id: 5/5